



An Augmented Reality Display for Conditionally Automated Driving

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Abstract

This paper investigates whether an Augmented Reality Head-up Display (AR-HUD) supports usability and reduces visual demand during conditionally automated driving. In a driving simulator study, 24 drivers experienced several driving scenarios while driving with conditional automation. The drivers completed one

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drive with a fully developed HMI designed for automated driving (AD-HMI) that presented visual information in the cluster display and included auditory and tactile output. In another drive, the same drivers were additionally supported by dynamic and static visual feedback via an AR-HUD concept. The latter was preferred by more than 80% of the sample due to the higher information content and the possibility to leave the eyes on the road. Drivers rated the AR concept to be better understandable and more useful. Eye-tracking revealed lower percentage of gazes to the instrument cluster during AR-HUD drives.

Author Keywords

Augmented reality; conditional automation; head-up-display;

ACM Classification Keywords

H.5.2. Information interfaces and presentation: Graphical user interfaces (GUI).

Introduction

Using systems defined as conditional automation [7] allows drivers to engage in non-driving related tasks (NDRTs) such as reading or watching videos. However, the driver is still the fallback level of the system and must be able to take over manual control within a certain time buffer. Numerous research studies have shown that drivers are in need of support in these take-over situations, and that non-supportive and badly



Figure 1: Overview of “basic” AD-HMI (UC = Use case; ODD = Operational Design Domain, see [7]). Copyright of all pictures at Continental Automotive GmbH.

designed HMIs can impair take-over performance (for a review see [3]). This may be especially the case when the driver needs to retrieve information from a head-down display, perform a series of gazes to mentally integrate the information with the driving situation and then decide on the appropriate action. But also in non-critical situations (e.g., automated lane changes) that likely create a need for information, head-down displays might go along with a higher visual demand. Thus, one approach for supporting the driver in both critical and non-critical driving situations could be the use of Augmented Reality Head-Up Displays (AR-HUDs) that allow drivers to stay focused on the driving situation. While several studies have applied AR in automated driving applications [4-6], there are no studies available that compare a fully-developed HMI using a head-down display with a version of the same HMI with enhanced AR-elements. In this study, we investigate whether such an AR-HUD increases the perceived usability of the HMI or lowers it as the AR-HUD could also be perceived as distracting from the driving task by adding visual elements to the driving scene.

Method

Study design

24 drivers experienced several driving scenarios while driving with conditional automation in a driving simulator. The drivers completed one drive with a fully developed HMI designed for automated driving (AD-HMI), including information in the instrument cluster, ambient LEDs, as well as acoustic and haptic feedback (see Figure 1 and 2). In another drive, the same drivers were additionally supported by dynamic and static visual feedback via an AR-HUD concept. The

order of drives was counter-balanced and treated as a between-subject factor.

Test course and non-driving related task

The study took place in the motion-based driving simulator of the WIVW GmbH using SILAB. The automated driving system maintained a speed of 120 km/h and adapted to speed limits. The automation was operated by an interaction device in the centre console.



Figure 2: Overview of HMI elements used in the simulator study.

One trial consisted of a 30 minutes-drive on a two-lane highway, including free driving sections, car following, experiencing a construction site the AD system was able to manage automatically and two lane changes where slower vehicles on the ego lane had to be overtaken. In-between these automated driving situations, a total of four take-over situations in which the system requested the driver to take over manual control were experienced. Two were issued due to construction sites where the ego-lane was blocked. The other two were triggered before broken-down vehicles positioned behind a curvy hill. At the moment of the

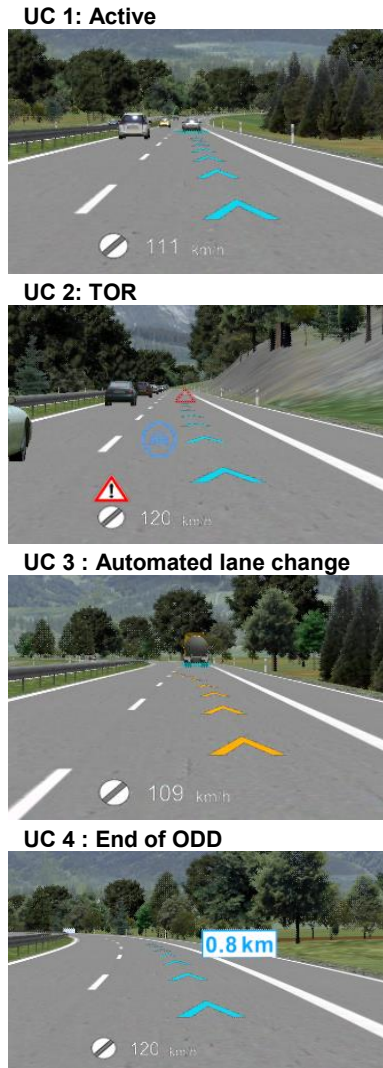


Figure 3: AR-HUD overview.

take-over request (TOR) the reason for the take-over could not yet be seen. A visually demanding non-driving related task (NDRT) was introduced which should distract the drivers in these situations. It consisted of a sequence of letters and figures displayed on a screen in a lower position of the centre console (so-called RSVP, see [1]). The driver had to react only to figures by pressing a button.

HMI

The “basic” AD-HMI (automated driving HMI, see Figure 2) consisted of (1) a display in the instrument cluster, (2) an LED strip below the windshield, (3) several acoustic messages as well as (4) haptic feedback on the seat for imminent warnings. During driving with the active system continuous information on the detected lanes, current speed and surrounding traffic was presented in the centre of the instrument cluster display (see Figure 1 row 1). In the left area, detected speed limits were displayed. In the right area, scheduled take-over requests were announced (see Figure 1 row 2). Automated lane changes were also displayed in the instrument cluster (see Figure 1 row 3). In case of a system limit a TOR was initiated 24 seconds before the situation was reached (see Figure 1 row 4) using a circle counting down time, a respective text message (“please take over the availability of automated driving will end”) and a sound melody with different escalation steps for increasing urgency.

The AR-HUD HMI variant consisted of information presented in the windshield in addition to the elements of the AD-HMI. Static information about speed, current speed limitation and reason for TORs or road-based regulations of the AD was presented in a conventional HUD display. Dynamic information about the trajectory

of the automated drive, as well as a marking of the objects relevant to collision avoidance and safely performing the take-over were presented dynamically in the driving scene through augmented reality. The dynamic AR elements were part of the driving scene and placed in relation to relevant objects. They consisted of a continuous path element on the road (so-called “fish-bones”; Figure 3 row 1) indicating which trajectory the system planned to perform as long as the system was active which was supposed to be helpful specifically in the situation when the automation keeps control even if it is somewhat not expected (such as automatic passing of construction site). In the latter case, we expected that awareness of the AD-mode would be achieved with lower visual workload with the AR-Display. In the take-over scenarios the take-over icon (similar to the one shown in the instrument cluster) was additionally displayed on the road. In take-over scenarios with the broken-down vehicle, a red triangle was placed on the lane and moved together with the vehicle until it stopped directly in front of the obstacle. This red triangle remained on the road even when the driver had already taken over the driving task and approached the situation in manual mode (Figure 3 row 2). The automated vehicles trajectory during automated lane changes as well as the remaining time until the end of the ODD (Operational Design Domain, see [7]) were also displayed in the AR-HUD (Figure 3, rows 3 and 4).

Dependent measures

After each of the two drives, the drivers rated the perceived understandability, usefulness and visual workload of the HMI feedback on a 16-point scale (see [2]) and gave a preference rating. As objective measures of visual workload, the percentage of gazes

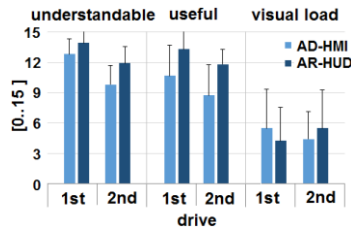


Figure 4: Global rating of HMI concept on understandability, usefulness and visual load.

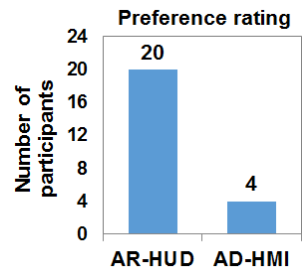


Figure 5: Preference rating after both drives.

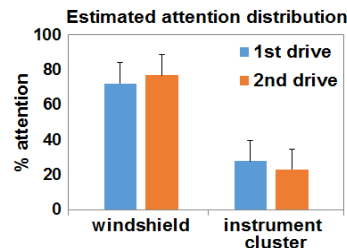


Figure 6: Attention distribution estimated by the drivers during AR-HUD drives.

towards the instrument cluster were analysed using a four-camera system from SmartEye. In the AR-HUD drive only, the drivers were also asked to self-assess how much attention they paid to the instrument cluster and to the windshield (where HUD is shown).

Results

Self-reported understandability, usefulness and visual workload were assessed after each drive (see Figure 4; error bars reflect standard deviation). The self-reported visual load was generally very low for both HMI variants and was on a comparable level (no significant effect of "HMI variant" $F[1;22]=.023$; $p=.881$ and "order of drives" $F[1;22]=.122$; $p=.949$). However, additional AR is perceived as more understandable ($F[1;22]=22.123$; $p<.001$) and useful ($F[1;22]=16.943$; $p<.001$). The factor "order of drive" was significant in both variables indicating a higher understandability ($F[1;22]=19.355$; $p<.001$) and usefulness ($F[1;22]=5.261$; $p=.032$) in the first drive. The interaction effect was not significant (understandable: $F[1;22]=2.206$; $p=.152$; useful: $F[1;22]=.089$; $p=.768$). After having perceived both drives with the different HMI variants 83% of the drivers (20 of the 24 participants) preferred the AR-HUD (see Figure 5). As reasons most participants argued that it a) makes it easier to perceive the relevant information as it is in the field of view and their attention can stay on the driving and b) it contains additional information which is especially useful in take-over situations. As apparent from Figure 7, the AR relieved the driver of the need to look down on the display as the fraction of this time substantially decreases with the AR-concept (significant main effect of "HMI variant": ($F[1;19]=13.332$, $p=.002$; no significant effect of "order of drive":

$F(1;19)=2.191$; $p=.155$). This effect was even more pronounced if the AR-HMI drive was the second one (significant interaction: $F[1;19]=7.494$; $p=.013$). Thus, it appears the AR-elements indeed support drivers in keeping attention at the driving scene. Drivers estimated that during AR-HUD drives about 75% of their attention was directed to information on the windshield and 25% at instrument cluster (see Figure 6), which reflects the objective data. There was no significant effect of "order of drives" ($t=-.849$; $df=17$; $p=.408$); drive 1: $M=72.3\%$; $SD=11.7\%$, drive 2: $M=76.9\%$, $SD=11.6\%$).

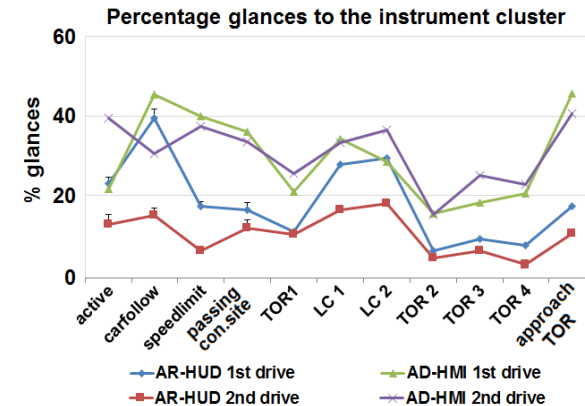


Figure 7: Percentage of glances to instrument cluster.

Discussion

Altogether it can be stated that the additional AR-HUD as supplement to the AD-HMI is highly accepted and appreciated for automated driving systems. It is preferred against the current AD-HMI, which is, however, already perceived as very understandable and useful. However, a more comprehensive analysis is needed, taking into account the driver's objective take-over performance.

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